

Humic Planosols:

Humic Planosols are tropical soils with a darker, humus-rich surface horizon that is periodically waterlogged, abruptly overlying a dense, clayey subsoil.

They are named “Humic” because of the relatively high organic matter content in the surface horizon, which increases nutrient-holding capacity and darkens the soil compared to Eutric and Ferralic Planosols. These soils typically occur on flat or gently undulating terrain in seasonally wet areas.

Drainage:

- **Wet season:** Surface horizon may stagnate above dense subsoil, but organic matter improves infiltration and structure.
- **Dry season:** Surface horizon firm and friable; subsoil slowly permeable but workable.
- **Landscape:** Flat to gently undulating terrain; often in low-lying areas or basin margins.

Depth:

- **Surface horizon:** 15–30 cm, sandy or loamy with high organic matter.
- **Subsoil:** Dense clay layer (>50 cm) with abrupt textural change; rooting may be slowed by subsoil density.

Workability:

- **When wet:** Topsoil sticky but manageable; better handling than Ferralic Planosols.
- **When dry:** Surface firm and friable; subsoil dense but workable; tillage easier with proper timing.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Moderate (15–25 cmol(+)/kg).
- **Base saturation:** Moderate (>35%), improved by organic matter.
- High organic matter content in the surface horizon enhances nutrient-holding capacity.
- Nitrogen and potassium moderate; phosphorus may be limited by fixation. Iron oxides present; micronutrients generally sufficient.

pH:

Slightly acidic to near neutral (5.0–6.5).

Management:

- Improve drainage via raised beds, ridges, or contouring.
- Incorporate organic matter (manure, compost) to enhance fertility, structure, and water retention.
- Apply phosphorus fertilizers carefully: slow-release rock phosphate combined with small doses of superphosphate.
- Avoid heavy machinery during wet periods to prevent compaction.
- Suitable for wetland rice, fodder crops, and other crops benefiting from higher fertility compared to Ferralic Planosols.